24/03/26

Ex: No: 07   IPC (Shared Memory)

Aim:

To implement Inter-Process Communication (IPC) using shared memory in C for data transfer between sender and receiver processes.

Sender code:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>
int main() {
    key_t key = 1234;
    int shmid;
    shmid = shmget (key, 1024, 0666 | IPC_CREAT);
    char *str = (char*) shmat (shmid, (void*)0, 0);
    printf("Enter message: ");
    scanf ("%[^\n]", str);
    printf("Data written to shared memory: %s\n", str);
    return 0;
}
```

Receiver code:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
int main() {
    key_t key = 1234;
    int shmid;
    shmid = shmget (key, 1024, 0666);
```

```c
    char *str = (char*) shmat (shmid, (void*)0, 0);
    printf ("Data read from shared memory: %s \n", str);
    shmdt (str);
    return 0;
}
```

Output1: (sender)

Enter message: IPC (Shared Memory)

Data written to shared memory: IPC (Shared Memory)

Output 2: (receiver)

Data read from shared memory: IPC (Shared Memory)

Result:

The sender writes data into shared memory and the receiver successfully reads the same data, demonstrating efficient communication between processes.